SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 1 – Analytical Problem solving (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO2	**Assessment:**	Assessment 1: Analytical Problem solving
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO2 Statement:	***Apply image enhancement, filtering, and segmentation techniques to analyze images, detect edges and boundaries, and solve real-world Computer Vision problems.***		
Student Name:	**NIVEDITHA**	**Reg. No.:**	192421440L
Section / Batch:	SLOT :A /2024	**Date:**	

Code of Conduct

I **NIVEDITHA/ 192421440L** *(NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.* Signature of the Student: **NIVEDITHA**

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Analytical

Problems

Q1. A grayscale image contains the following pixel intensity values: 10, 10, 20, 20, 20, 30, 30, and 40. Construct the histogram by calculating the frequency of each intensity value. Calculate the cumulative distribution function (CDF) for the given image. Apply histogram equalization using the formula s = round((L − 1) × CDF), where L
= 256. Determine the new intensity values and compare the contrast distribution before and after equalization.

ANS:

Histogram Equalization

1. Given Pixel Intensity Values

- Given grayscale pixel values: 10, 10, 20, 20, 20, 30, 30, 40.
- Total number of pixels = 8.
- Number of intensity levels $L=256$.
- Histogram equalization is an image enhancement technique.
- It improves contrast by spreading intensity values over a wider range.

2. Histogram Calculation

- Intensity 10 occurs 2 times.
- Intensity 20 occurs 3 times.
- Intensity 30 occurs 2 times.
- Intensity 40 occurs 1 time.
- The total frequency is $2+3+2+1=8$.
- These frequencies represent the distribution of pixel intensities.

3. Cumulative Distribution Function (CDF)

- CDF is calculated by adding the frequencies up to each intensity and dividing by the total number of pixels.
- For intensity 10: $CDF=2/8=0.25$.
- For intensity 20: $CDF=5/8=0.625$.

- For intensity 30: \(CDF=7/8=0.875\).
- For intensity 40: \(CDF=8/8=1.0\).

4. Histogram Equalization Calculation

- Formula: \(s=\text{round}((L-1)\times CDF)\).
- For intensity 10: \(255\times0.25=63.75\approx64\).
- For intensity 20: \(255\times0.625=159.375\approx159\).
- For intensity 30: \(255\times0.875=223.125\approx223\).
- For intensity 40: \(255\times1=255\).

5. New Intensity Values

- 10 → 64.
- 20 → 159.
- 30 → 223.
- 40 → 255.
- Equalized pixel values: 64, 64, 159, 159, 159, 223, 223, 255.

6. Comparison and Advantages

- Before equalization, intensity values are concentrated between 10 and 40.
- After equalization, the values spread across a wider intensity range.
- The difference between intensity levels becomes more noticeable.
- It improves image contrast and visibility.
- It helps reveal details in low-contrast images.
- It is useful in medical imaging, satellite images, and low-light photographs.

Q2. A 3 × 3 image has the intensity values [[10, 20, 30], [20, 100, 20], [10, 20, 30]]. Apply a 3 × 3 averaging filter to calculate the new intensity value of the centre pixel. Replace the centre pixel with the calculated average while keeping the other pixels unchanged. Apply a median filter to the same neighbourhood and calculate the centre pixel value. Compare the results of the averaging and median filters and determine which filter better preserves image details in the presence of noise.

ANS:

Averaging Filter and Median Filter

1. Given Image

- Given 3 × 3 image:

\[\begin{bmatrix} 10&20&30\\ 20&100&20\\ 10&20&30 \end{bmatrix} \]

- Centre pixel = 100.
- The averaging and median filters are used to reduce noise.
- Both filters process the neighbourhood around the centre pixel.

2. Averaging Filter

- The averaging filter calculates the mean of all pixels in the 3 × 3 neighbourhood.
- Formula: Average = Sum of pixel values / Total number of pixels.
- Sum = \(10+20+30+20+100+20+10+20+30=260\).
- Total number of pixels = 9.
- Average = \(260/9=28.89\approx29\).
- New centre pixel = 29.

3. Result After Averaging Filter

\[\begin{bmatrix} 10&20&30\\ 20&29&20\\ 10&20&30 \end{bmatrix} \]

- The centre value changes from 100 to 29.
- The filter smooths the image by reducing differences between neighbouring pixels.
- It reduces some types of noise.
- However, it may blur edges and important details.

4. Median Filter

- The median filter arranges all pixel values in ascending order.
- Sorted values: 10, 10, 20, 20, 20, 20, 30, 30, 100.

- Total number of values = 9.
- The middle value is the fifth value.
- Median = 20.
- New centre pixel = 20.

5. Result After Median Filter

\[\begin{bmatrix} 10&20&30\\ 20&20&20\\ 10&20&30 \end{bmatrix} \]

- The centre pixel changes from 100 to 20.
- The median filter removes an unusually high value from the neighbourhood.
- It is effective for salt-and-pepper noise.
- It usually preserves important edges better than averaging.

6. Comparison and Conclusion

- Averaging filter uses the mean of all pixel values.
- Median filter uses the middle value after sorting.
- Averaging smooths the image but may blur boundaries.
- Median filtering removes outliers while preserving edges more effectively.
- Both filters can improve image quality.
- Conclusion: The median filter is more suitable when noise must be removed without losing important image details.

Q3. Consider a grayscale image with the intensity matrix [[50, 50, 50], [50, 100, 50], [50, 50, 50]]. Apply the

Laplacian filter mask [[0, −1, 0], [−1, 4, −1], [0, −1, 0]] to calculate the response at the centre pixel. Use the calculated response to sharpen the centre pixel by adding it to the original intensity value. Determine the resulting centre pixel intensity after sharpening. Explain how the Laplacian filter enhances intensity transitions and fine details in the image.

ANS:

Image Sharpening Using Laplacian Filter

1. Given Image

- Given grayscale matrix:

$$\begin{bmatrix} 50&50&50\\ 50&100&50\\ 50&50&50 \end{bmatrix}$$

- Original centre pixel = 100.
- The Laplacian filter is used to detect rapid changes in pixel intensity.
- It helps enhance edges and fine details.

2. Given Laplacian Mask

$$\begin{bmatrix} 0&-1&0\\ -1&4&-1\\ 0&-1&0 \end{bmatrix}$$

- The centre coefficient is 4.
- The four neighbouring coefficients are −1.
- The mask compares the centre pixel with its neighbouring pixels.
- A large response indicates a significant intensity change.

3. Calculate Laplacian Response

- Formula: Response = $4\times\text{Centre}-\text{Sum of four neighbours}$.
- Centre pixel = 100.
- Neighbouring pixels = 50, 50, 50, 50.
- Response = $4(100)-(50+50+50+50)$.
- Response = $400-200$.
- Laplacian response = 200.

4. Sharpen the Centre Pixel

- Formula: New pixel = Original pixel + Laplacian response.
- Original centre pixel = 100.

- Laplacian response = 200.
- New pixel = \(100+200\).
- Resulting centre pixel = 300.
- Since an 8-bit image supports values from 0 to 255, the displayed value is clipped to 255.

5. Explanation

- The Laplacian filter detects changes between neighbouring pixels.
- It produces a strong response where intensity changes rapidly.
- Adding the response to the original image enhances the centre pixel.
- This increases the difference between the centre and its surroundings.
- As a result, edges and fine details become more visible.
- Flat regions with similar intensity values generally produce a smaller response.

6. Advantages and Applications

- Improves image sharpness.
- Highlights object boundaries.
- Makes fine details clearer.
- Supports edge detection and image analysis.
- It is useful in medical images, document processing, and object recognition.
- Excessive sharpening may increase noise or create overly bright pixels.

Q4. A grayscale image contains the pixel intensity values [[40, 60, 120], [80, 150, 200], [30, 100, 180]]. Apply global thresholding with a threshold value of 100, assigning 1 to pixels greater than or equal to the threshold and 0 to the remaining pixels. Construct the resulting binary image. Calculate the number of foreground and background pixels after thresholding. Explain how the selected threshold separates objects from the background and identify one limitation of this method.

ANS:

Global Thresholding

1. Given Image

- Given grayscale matrix:

\[\begin{bmatrix} 40&60&120\\ 80&150&200\\ 30&100&180 \end{bmatrix} \]

- Threshold value = 100.
- Global thresholding converts a grayscale image into a binary image.
- It separates foreground and background using one fixed threshold value.

2. Thresholding Rule

- If pixel intensity is greater than or equal to 100, assign 1.
- If pixel intensity is less than 100, assign 0.
- Foreground pixels are represented by 1.
- Background pixels are represented by 0.
- Each pixel is checked against the threshold value.

3. Resulting Binary Image

Binary Image

0

0

1

0

1

1

0

1

1

Foreground (1)

Background (0)

4. Foreground Pixel Calculation

- Pixels greater than or equal to 100 are 120, 150, 200, 100, and 180.
- Total foreground pixels = 5.
- These pixels are assigned the value 1.
- They represent the selected object or bright region.

5. Background Pixel Calculation

- Pixels below 100 are 40, 60, 80, and 30.
- Total background pixels = 4.
- These pixels are assigned the value 0.
- Total foreground and background pixels = \(5+4=9\).

6. Explanation

- Global thresholding is simple and computationally efficient.
- It separates pixels based on their intensity values.
- It simplifies the image for further processing.
- It is useful for separating text from paper or objects from backgrounds.
- The selected threshold affects the final segmentation result.

7. Limitation and Applications

- A single threshold may not work well under uneven lighting.
- Shadows can cause incorrect classification.
- Objects with intensity similar to the background may not be separated correctly.
- Adaptive thresholding can be used when lighting varies across an image.
- Global thresholding is useful in document scanning, basic object detection, and image segmentation.

Q5. A grayscale image has the intensity matrix [[10, 10, 10], [10, 50, 90], [10, 50, 90]]. Apply the Sobel operator

using the masks Gx = [[−1, 0, 1], [−2, 0, 2], [−1, 0, 1]] and Gy = [[−1, −2, −1], [0, 0, 0], [1, 2, 1]] to calculate the gradient components at the centre pixel. Calculate the gradient magnitude using $G = \sqrt{(Gx^2 + Gy^2)}$. Determine the dominant edge direction based on the calculated gradient components. Explain how the detected edge information can be used for boundary detection and image segmentation.

ANS:

Edge Detection Using the Sobel Operator

1. Given Image

$$\begin{bmatrix} 10&10&10\\ 10&50&90\\ 10&50&90 \end{bmatrix}$$

- The Sobel operator detects changes in image intensity.
- It calculates gradients in horizontal and vertical directions.
- It helps identify edges, shapes, and object boundaries.

2. Sobel Masks

Horizontal gradient mask:

$$G_x=\begin{bmatrix} -1&0&1\\ -2&0&2\\ -1&0&1 \end{bmatrix}$$

Vertical gradient mask:

$$G_y=\begin{bmatrix} -1&-2&-1\\ 0&0&0\\ 1&2&1 \end{bmatrix}$$

- G_x measures intensity changes in the horizontal direction.
- G_y measures intensity changes in the vertical direction.
- Larger gradient values indicate stronger intensity changes.

3. Calculate Horizontal Gradient G_x

- Top row contribution = $(-1\times10)+(1\times10)=0$.
- Middle row contribution = $(-2\times10)+(2\times90)=160$.
- Bottom row contribution = $(-1\times10)+(1\times90)=80$.
- Total = $0+160+80$.
- Horizontal gradient: $G_x=240$.

4. Calculate Vertical Gradient G_y

- Top row contribution = $(-1\times10)+(-2\times10)+(-1\times10)=-40$.

- Middle row contribution = 0.
- Bottom row contribution = \((1\times10)+(2\times50)+(1\times90)=200\).
- Total = \(-40+0+200\).
- Vertical gradient: \(G_y=160\).

5. Calculate Gradient Magnitude

- Gradient magnitude represents the strength of an edge.
- Formula: \(G=\sqrt{G_x^2+G_y^2}\).
- Substitute the calculated values:

\[G=\sqrt{240^2+160^2} \]

\[G=\sqrt{57600+25600} \]

\[G=\sqrt{83200} \]

- Gradient magnitude ≈ 288.44.
- A large magnitude indicates a strong intensity change at the centre.

6. Determine Dominant Edge Direction

- Horizontal gradient = 240.
- Vertical gradient = 160.
- The horizontal intensity change is stronger than the vertical intensity change.
- Gradient direction is approximately \(33.7^\circ\) from the horizontal.
- The edge orientation is perpendicular to the gradient, approximately \(123.7^\circ\), or equivalently \(-56.3^\circ\).
- This indicates a strong edge with a greater horizontal intensity change.

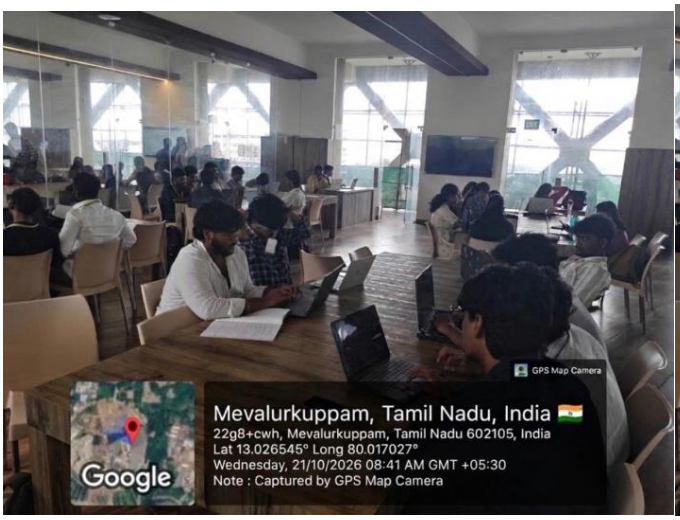
7. Boundary Detection and Segmentation

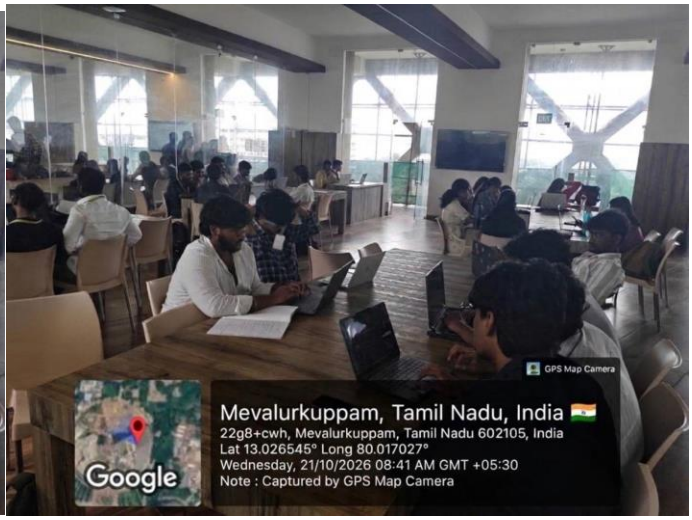
- Sobel detects edges by identifying sudden changes in pixel intensity.
- Edge information helps locate the boundaries of objects.
- The detected edges can be combined to outline an object.
- Segmentation separates the object from the background.
- It helps identify shapes and regions for further analysis.

8. Advantages and Applications

- Detects object boundaries and edges.
- Helps separate objects from backgrounds.

- Supports image segmentation and object recognition.
- Useful in medical imaging and automated inspection.
- Helps identify shapes and outlines in photographs.
- It is simple to implement, but noise may create false edges.





This Assessment tool has been discussed in the class and the tasks has been successfully done on 21/10/2026.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____